

Emotional Avatars: Effect of Uncanniness in Identifying Emotions using Avatar Expressions

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ABSTRACT

This work analyses the effect of uncanniness behaviour in identifying emotions from different humanoid avatar representations. Expressive avatars play a vital role in immersive environments. However, technical limitations in replicating subtle emotional cues using real-time expression conversion techniques create an uncanniness to the viewers. Hence, achieving the desired emotional awareness is arguable. Therefore, using an avatar representation resistant to uncanniness in systems sensitive to emotional changes is vital. Therefore, here we analyse the level of uncanniness noticed by people for different avatars exhibiting various emotions using expressive faces and the behavioural trends of people in catching emotion-related uncanniness.

Keywords: Emotional Avatars, Emotion Sensitive Systems, Emotion Related Uncanniness

1 INTRODUCTION

Non-verbal communication intelligence is vital to establish, understand and maintain social relationships and collaborations. According to Ekman [3], emotions occur when we sense, rightly or wrongly, that something that might seriously affect our welfare for better or worse is happening or about to happen. Most of the time, emotional facial expressions happen without our consciousness; throughout evolution, it has been helpful to help others sense danger and take action. Hence, in situations like counselling sessions, student-teacher communication, angry customer handling, criminal investigations, it is crucial to identify those expressions or micro-expressions that happen naturally according to the given scenario [2, 6, 7, 10].

With the rapid advancement of social interaction platforms that use virtual reality and mixed reality technologies, most of the above communications and collaborations also converted into digital forms using different avatar types. Real-time expressions reenactment technologies are used to reproduce the facial expressions on avatars to provide realistic interactions in these systems [4, 5].

However, with the technical limitations, some delicate facial expressions responsible for revealing emotional cues are not adequately reenacted on the avatar faces. This situation creates an imbalance between the avatar physical realism and behaviour realism which might generate an uncanniness that affects the avatar emotion identification capability [1].

This dilemma raises the concern of finding the proper avatar representation to use in real-time collaborative platforms to support the user's emotional communication [9].

Hence, this paper discusses the level of uncanniness people perceive from different avatars and the emotion identification capability of these avatar representations regarding eight basic emotions. The

methodology used, results from analysis and discussion of results are presented in the latter part of this paper.

2 METHODOLOGY AND STUDY DESIGN

We conducted a study and analyze the effect of uncanniness in reading dynamic emotional expression from three avatar/actor types (Real human, Photorealistic avatar, Cartoonized avatar). Figure 1 shows the selected avatar/actor set. Accordingly, we attempted to answer the below research question by analysing these hypotheses.

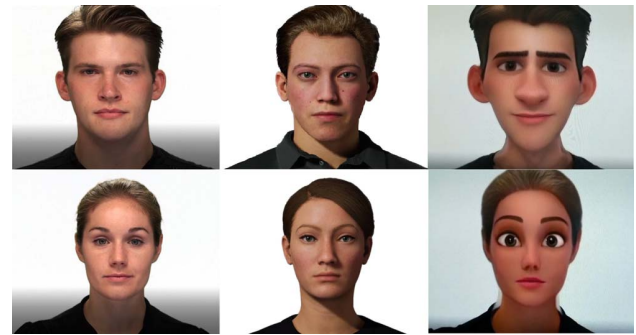


Figure 1: Real Human, Photorealistic and Cartoonized Avatar counterparts

- How do different emotional-avatar representations (Video of real humans vs. Photorealistic vs. Cartoonized) affect our perceptions of uncanniness?
 - H_1 : People report uncanniness at different rates when observing emotions through Real-human avatars, Photorealistic avatars and Cartoonized avatars.
 - H_2 : The ability for people to correctly identify emotions will vary depending on the avatar representation (Real-human vs. Photorealistic vs. Cartoonized).

First, "Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)" was selected as the reference dataset to replicate validated emotional representations on avatars [8]. Hence one male and one female actor from the dataset was picked, and their counterpart photorealistic¹, and cartoonized² avatars were created. Next, the emotional expressions of the respective avatars were reenacted on the photorealistic and cartoonized avatars using a real-time facial expression reenactment software package³. RAVDESS dataset has eight different emotion categories (neutral, calm, happy, sad, angry, fearful, disgust and surprise). Therefore we ended up with three video sets consisting of sixteen videos in each set (8 male expressions, 8 female).

¹<https://metahuman.unrealengine.com/>

²<https://www.snapchat.com/>

³<https://vimeo.com/549059633>

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During the study, we presented the three video sets to the participants in the order of real human, photorealistic avatar and cartoonized avatar. First, participants were asked to identify the emotion represented by each video from the given eight options. Next, at the end of each video set, a questionnaire related to the uncanniness of the dynamic emotional representation was answered related to each avatar by the participants.

3 RESULTS

40 participants (22 male, 18 female; age $M = 27.2$, $SD = 6.77$) were involved in this study.

3.0.1 Hypothesis H_1 Testing

The dependent variable of uncanniness was tested against the independent variable of avatar type for this hypothesis. Our finding supported the null hypothesis, and we conducted an epsilon squared test(0.45) according to a Kruskal-Wallis test ($p < 0.001$). Since there was a "large" effect, we conducted pairwise comparisons using a Wilcoxon-Rank Sum test, and the results are presented in Table 1.

Parameter	Group 1	Group 2	n1	n2	p
Uncanniness	Cartoonized	Photorealistic	40	40	<0.001
Uncanniness	Cartoonized	Real.Human	40	40	=0.788
Uncanniness	Photorealistic	Real.Human	40	40	<0.001

Table 1: Pairwise Comparison for Reported Uncanniness by Avatar Type.

3.0.2 Hypothesis H_2 Testing

The dependent variable of emotion accuracy was calculated against the independent variable avatar type in testing this hypothesis. First, we ran the Kruskal-Wallis test ($p < 0.001$) which showed us there is a considerable difference between the accuracy for different avatar types. Since there is a considerable difference, we conducted a Kruskal effect size test ($y = accuracy$, $n = 240$, $effectsize = 0.184$, $magnitude = large$). Next, we conducted a Wilcoxon test to find the emotion identification accuracy differences between the avatar type groups. Results are reported in Table 2. Then we tested the emotion identification accuracy by testing against participant gender. For that, we have again conducted a Kruskal-Wallis test but found no significant difference ($p = 0.060$).

Parameter	Group 1	Group 2	n1	n2	p
Accuracy	Cartoonized	Photorealistic	80	80	<0.001
Accuracy	Cartoonized	Real.Human	80	80	=0.089
Accuracy	Photorealistic	Real.Human	80	80	<0.001

Table 2: Wilcoxon test of Accuracy by Avatar Type.

4 CONCLUSION

Avatar type carries a significant impact on creating an uncanny feeling towards emotional expressions for users. Photorealistic avatars have the highest level of uncanniness related to these dynamic emotional expressions, while cartoonized avatars do not generate any difference from real humans. Apart from that, these uncanny emotional expressions also affect the user's emotion identification capability. Photorealistic avatar emotional expressions were the most challenging to identify, while cartoonized avatar expressions deliver the same expression explanation level as real humans.

Real-time facial expression mapping techniques evolve rapidly and have a great potential in fueling socially interactive applications

for the better. However, those techniques can still not accurately map the subtle cues important for emotional expression communication. Hence, users sense uncanniness from those expressions, which lead to inaccurate emotion identifications. Thus, these less accurate expressions could affect the effectiveness of these interactive systems by devaluing the quality of communication. Furthermore, this could significantly harm the collaboration of the systems designed to establish and maintain relationships based on emotional intelligence.

Hence, this study emphasises the importance of selecting the appropriate avatar type for real-time dynamic expression mapping systems to ensure accurate expression communication.

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